

EE/CE 252/251

Signals and Systems

Section L3

Problem

Find $X[z]$

$$x[n] = 3\delta[n - 3] + \delta[n - 5] - \delta[n - 7] - 3u[n - 9].$$

Problem

Find $X[z]$

$$x[n] = 3\delta[n-3] + \delta[n-5] - \delta[n-7] - 3u[n-9].$$

Using
Table we
get

$$X[z] = \underbrace{3z^{-3} + z^{-5} - z^{-7}}_{|z|>0} - \underbrace{3z^{-9} \frac{z}{z-1}}_{|z|>1}.$$

$$X[z] = 3z^{-3} + z^{-5} - z^{-7} - \frac{3z^{-8}}{z-1}, \quad \text{ROC: } |z| > 1.$$

Problem

Using the definition of the z -transform, find the z -transform and the ROC for each of the following signals.

- (a) $u[n - m]$
- (b) $\gamma^n \sin \pi n u[n]$
- (c) $\gamma^n \cos \pi n u[n]$

$\sin(\pi n)$



(f) $\sum_{k=0}^{\infty} 2^{2k} \delta[n - 2k]$

(k) $[2^{n-1} - (-2)^{n-1}]u[n]$

(a)

$$\begin{aligned} X[z] &= \sum_{n=m}^{\infty} z^{-n} = z^{-m} + z^{-(m+1)} + z^{-(m+2)} + \dots \\ &= z^{-m} \left[1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right] \\ &= z^{-m} \left(\frac{1}{1 - \frac{1}{z}} \right) = \frac{z}{z^m(z-1)}, \quad |z| > 1. \end{aligned}$$

(b) Notice that $\gamma^n \sin \pi n u[n] = 0$ for all n . Hence,

$$X[z] = 0, \quad \text{all } z.$$

(c) Since $\gamma^n \cos \pi n u[n] = (-\gamma)^n u[n]$, we see that

$$X[z] = \frac{z}{z + \gamma}, \quad |z| > |\gamma|.$$

(f)

$$\sum_{k=0}^{\infty} 2^{2k} \delta[n - 2k] = \delta[n] + 4\delta[n - 2] + 16\delta[n - 4] + \dots$$

$$X[z] = 1 + \frac{4}{z^2} + \frac{16}{z^4} + \frac{64}{z^6} + \dots$$

This is a geometric progression with common ratio $\frac{4}{z^2}$. Hence,

$$X[z] = \frac{1}{1 - \frac{4}{z^2}} = \frac{z^2}{z^2 - 4} \quad |z| > 2.$$

(k)

$$x[n] = [2^{n-1} - (-2)^{n-1}] u[n]$$

$$\begin{aligned} X[z] &= \frac{1}{2} \left[\sum_{n=0}^{\infty} \left(\frac{2}{z} \right)^n + \sum_{n=0}^{\infty} \left(\frac{-2}{z} \right)^n \right] \\ &= \frac{1}{2} \left[\frac{1}{1 - \frac{2}{z}} + \frac{1}{1 + \frac{2}{z}} \right] \\ &= \frac{1}{2} \left[\frac{z}{z-2} + \frac{z}{z+2} \right] \\ &= \frac{z^2}{z^2 - 4}, \quad |z| > |2|. \end{aligned}$$

Problem

Using only the z -transforms of Table 5.1, determine the z -transform of each of the following signals.

(a) $u[n] - u[n - 2]$

(b) $\gamma^{n-2}u[n - 2]$

(c) $2^{n+1}u[n - 1] + e^{n-1}u[n]$

(d) $\left[2^{-n} \cos\left(\frac{\pi}{3}n\right)\right]u[n - 1]$

(e) $n\gamma^n u[n - 1]$

(a)

$$u[n] - u[n - 2] = \delta[n] + \delta[n - 1]$$

Hence,

$$u[n] - u[n - 2] \iff 1 + \frac{1}{z} = \frac{z + 1}{z}.$$

(b)

$$\gamma^{n-2}u[n-2] = \frac{1}{\gamma^2} \{ \gamma^n u[n] - \delta[n] - \gamma \delta[n-1] \}$$

Hence,

$$\gamma^{n-2}u[n-2] \iff \frac{1}{\gamma^2} \left[\frac{z}{z-\gamma} - 1 - \frac{\gamma}{z} \right] = \frac{1}{z(z-\gamma)}.$$

(c)

$$x[n] = (2)^{n+1}u[n-1] + (e)^{n-1}u[n] = 4(2)^{n-1}u[n-1] + \frac{1}{e}(e)^nu[n]$$

Therefore,

$$X[z] = \frac{4}{z-2} + \frac{1}{e} \frac{z}{z-e}.$$

(d)

$$x[n] = \left[(2)^{-n} \cos \frac{\pi n}{3} \right] u[n-1] = (2)^{-n} \cos \frac{\pi n}{3} u[n] - \delta[n]$$

Therefore,

$$X[z] = \frac{z(z-0.25)}{z^2-0.5z+0.25} - 1 = \frac{0.25(z-1)}{z^2-0.5z+0.25}.$$

Problem

Given $X[z]$ find $x[n]$

$$(a) \quad \frac{z(z-4)}{z^2-5z+6}$$

$$(e) \quad \frac{z(2z+3)}{(z-1)(z^2-5z+6)}$$

$$(f) \quad \frac{z(-5z+22)}{(z+1)(z-2)^2}$$

(a)

$$\frac{X[z]}{z} = \frac{z-4}{(z-2)(z-3)} = \frac{2}{z-2} - \frac{1}{z-3}$$

$$X[z] = 2\frac{z}{z-2} - \frac{z}{z-3}$$

$$x[n] = [2(2)^n - (3)^n] u[n]$$

(e)

$$\frac{X[z]}{z} = \frac{2z+3}{(z-1)(z-2)(z-3)} = \frac{5/2}{z-1} - \frac{7}{z-2} + \frac{9/2}{z-3}$$

$$X[z] = \frac{5}{2}\frac{z}{z-1} - 7\frac{z}{z-2} + \frac{9}{2}\frac{z}{z-3}$$

$$x[n] = \left[\frac{5}{2} - 7(2)^n + \frac{9}{2}(3)^n \right] u[n]$$

(f)

$$\frac{X[z]}{z} = \frac{-5z + 22}{(z + 1)(z - 2)^2} = \frac{3}{z + 1} + \frac{k}{z - 2} + \frac{4}{(z - 2)^2}$$

Multiply both sides by z and let $z \rightarrow \infty$. This yields

$$0 = 3 + k + 0 \implies k = -3.$$

Thus,

$$X[z] = 3\frac{z}{z + 1} - 3\frac{z}{z - 2} + 4\frac{z}{(z - 2)^2}$$
$$x[n] = [3(-1)^n - 3(2)^n + 2n(2)^n] u[n].$$

We will use
L'Hopital rule to
resolve $\frac{\infty}{\infty}$ forms

Problem

Find the output $y[n]$ of an LTID system specified by the equation

$$\begin{aligned} 2y[n+2] - 3y[n+1] + y[n] \\ = 4x[n+2] - 3x[n+1] \end{aligned}$$

for input $x[n] = (4)^{-n}u[n]$ and initial conditions $y[-1] = 0$ and $y[-2] = 1$.

The system equation in delay form is

$$2y[n] - 3y[n-1] + y[n-2] = 4x[n] - 3x[n-1].$$

Also

$$\begin{aligned} y[n] &\Longleftrightarrow Y[z], & y[n-1] &\Longleftrightarrow \frac{1}{z}Y[z], & y[n-2] &\Longleftrightarrow \frac{1}{z^2}Y[z] + 1, \\ x[n] &\Longleftrightarrow X[z] = \frac{z}{z-0.25}, & \text{and} & & x[n-1] &\Longleftrightarrow \frac{1}{z-0.25}. \end{aligned}$$

The z -transform of the equation is

$$2Y[z] - \frac{3}{z}Y[z] + \frac{1}{z^2}Y[z] + 1 = \frac{4z}{z-0.25} - \frac{3}{z-0.25} = \frac{4z-3}{z-0.25}$$

or

$$\left(2 - \frac{3}{z} + \frac{1}{z^2}\right) Y[z] = -1 + \frac{4z-3}{z-0.25} = \frac{3z-2.75}{z-0.25}.$$

Thus,

$$\begin{aligned}\frac{Y[z]}{z} &= \frac{z(3z - 2.75)}{(2z^2 - 3z + 1)(z - 0.25)} \\ &= \frac{z(3z - 2.75)}{2(z - 0.5)(z - 1)(z - 0.25)} \\ &= \frac{5/2}{z - 1/2} + \frac{1/3}{z - 1} - \frac{4/3}{z - 0.25} \\ y[n] &= \left[\frac{1}{3} + \frac{5}{2}(0.5)^n - \frac{4}{3}(0.25)^n \right] u[n] \\ &= \left[\frac{1}{3} + \frac{5}{2}(2)^{-n} - \frac{4}{3}(4)^{-n} \right] u[n].\end{aligned}$$

Problem

For each impulse response, determine the number of system poles, whether the poles are real or complex, and whether the system is BIBO-stable.

(a) $h_1[n] = (-1 + (0.5)^n)u[n]$

$$H_1[z] = -\frac{z}{z-1} + \frac{z}{z-0.5}$$

- The system has two poles ($z = 1, z = 0.5$)
- One of these is not inside the unit circle.
- System is not BIBO stable.

Problem

A system has impulse response $h[n] = u[n - 3]$.

- (a) Determine the impulse response of the inverse system $h^{-1}[n]$.
- (b) Is the inverse stable? Is the inverse causal?

$$H[z] = z^{-3} \frac{z}{z-1},$$

- Only has one zero at $z = 0$ (inside the unit circle).
- Therefore the inverse exists and is stable.

$$H^{-1}[z] = \frac{1}{H[z]} = \frac{z-1}{z^{-2}} = z^3 - z^2.$$

$$h^{-1}[n] = \delta[n+3] - \delta[n+2].$$

We see that the inverse is also causal (is zero for negative values of n).